The University of Burdwan
Center & College: Chandernagore College (404)
B.Sc 3rd Semester Practical Examination, 2024(CBCS)

Couse Name: Data Structures Lab

Course Code: COMP 3012 (Practical)

FULL MARKS: 20

Subject: Computer Science

A. EXPERIMENT (ATTEMPT ANY 1 QUESTION)

Time: 2 Hours

MARKS: 10

1. Write a menu driven C program for single linked list to do the following operations:
   a. Create a list with *n* elements, where the value of *n* and the values of those elements are given by the user
   b. Display the list
   c. Insert a new node at the end of the list, value of the new node is given by the user
2. Write a menu driven C program for double linked list to do the following operations:
   a. Create a list with *n* elements, where the value of *n* and the values of those elements are given by the user
   b. Display the list
   c. Insert a new node at the begining of the list, value of the new node is given by the user
3. Write a menu driven C program to implement Stack using array to do the following operations:
   a. PUSH
   b. POP
   c. PEEK
4. Write a menu driven C program to implement Stack using linked list to do the following operations:
   a. PUSH
   b. POP
   c. PEEK
5. Write a menu driven C program to implement Queue using array to do the following operations:
   a. EN-QUEUE
   b. DE-QUEUE
   c. PEEK
6. Write a menu driven C program to implement Queue using linked list to do the following operations:
   a. EN-QUEUE
   b. DE-QUEUE
   c. PEEK
7. Write a menu driven C program to implement Binary Search Tree to do the following operations:
   a. Create a tree with *n* elements, where the value of *n* and the values of those elements are given by the user
   b. Display using In-Order Traversal
   c. Display using Post-Order Traversal

B. LAB NOTEBOOK — MARKS: 5

C. VIVA — MARKS: 5